

ab

•

$$L(a) = \{a\}$$

$$L(b) = \{b\}$$

$$L(ab) = \{ab\} \quad L(a)L(b) = \{a\}\{b\} = \{ab\}$$

a + b

$$L(a+b) = L(a) \cup L(b) = \{a\} \cup \{b\} = \{a, b\}$$

ab* = a(b*)

$$L(b^*) = L(b)^* = \{b\}^* = \{\epsilon, b, bb, \dots\}$$

$$L(a(b^*)) = L(a)L(b^*) = \{a\}\{\epsilon, b, bb, \dots\} = \{a, ab, abb, \dots\}$$

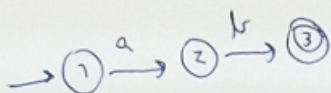
(ab)*

$$L((ab)^*) = L(ab)^* = \{ab\}^* = \{\epsilon, ab, abab, \dots\}$$

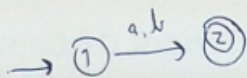
a + b* = a + (b*)

$$L(a + (b^*)) = L(a) \cup L(b^*) = \{a\} \cup \{\epsilon, b, bb, \dots\} = \{a, \epsilon, b, bb, \dots\}$$

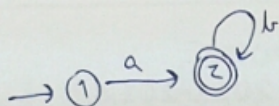
ab



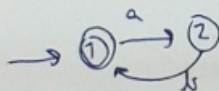
$a+b$



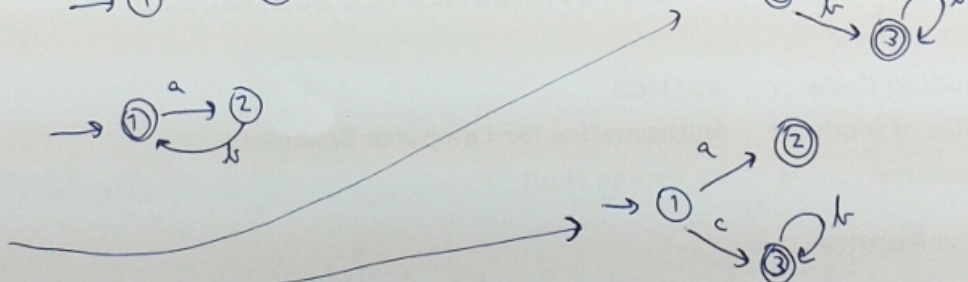
ab^*



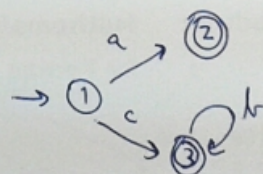
$(ab)^*$



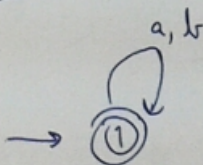
$a+b^*$



$a+cb^*$



$(a+b)^*$



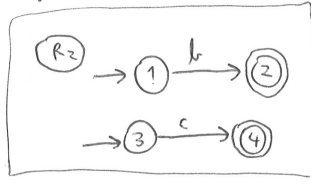
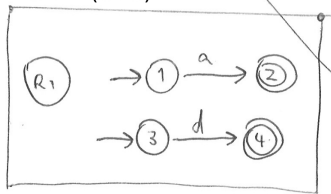
3. RE $\rightarrow$ NFA

convert R_5 into an NFA

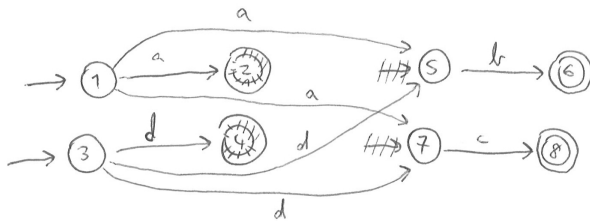
$$R_5 = R_2 (R_1 R_2)^*$$

$$R_2 = (b + c)$$

$R_1 = (a+d)$



$R_1 R_2$



clean up to

